

assignment #05

Course: CHE 341: CHEMICAL PROCESS
CONTROL

Term: Spring 2026

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Due Date: Friday, April 10th in class

AVIATION THEME

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Bring a hard copy of your report as a PDF file by the Due Date and Time(**Beginning of the Lecture**) in class.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Empirical Model Building

Construct the process transfer function from the multiple step test commercial data contained in the accompanying Excel file: **Assignment-05.xlsx**

Chemical Process Control

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To begin constructing the process transfer function, which will take on the FOPDT transfer function shown below, the following steps will be necessary.

$$G_p(s) = \frac{K_p e^{-\theta s}}{\tau s + 1}$$

For a step change in manipulated variable occurring at time t_s , the time-domain model response is

$$PV_{\text{model}}(t) = \begin{cases} PV_0, & t < t_s + \theta \\ PV_0 + K_p \Delta MV (1 - e^{-(t-t_s-\theta)/\tau}), & t \geq t_s + \theta \end{cases}$$

In this expression, PV_0 is the average process variable before the step and ΔMV is the size of the manipulated-variable step. This model assumes that the process remains constant until the dead time has passed, after which the response approaches the new steady value with first-order behavior.

First the dataset was divided into one fitting window and three additional disturbance windows. These windows were selected so that each one contained a clear manipulated-variable transition along with enough data before and after the step to define the baseline and the response. The selected windows are summarized below.

Disturbance	Time Window	Purpose
Fitting window	12:52:12 to 12:56:26	Initial parameter estimation
Downstep	12:56:28 to 13:00:00	Validation / separate disturbance fit
Upstep	13:00:40 to 13:04:52	Validation / separate disturbance fit
Downstep 2	13:04:52 to end	Validation / separate disturbance fit

For each disturbance, the following quantities were defined from the data:

u_0 = average MV before the step

u_f = average MV after the step

$\Delta MV = u_f - u_0$

PV_0 = average PV before the step

PV_f = average PV after the response has largely settled

$\Delta PV = PV_f - PV_0$

The process gain was initially estimated using

$$K_p \approx \frac{\Delta PV}{\Delta MV}$$

This gain estimate provided a physically reasonable starting value for the least-squares fit. The sign of ΔMV and ΔPV was preserved, so that upsteps and down steps were handled consistently.

Because the data was noisy and the manipulated-variable transitions were not perfectly instantaneous, the parameters were not determined only from a few characteristic points. Instead, each disturbance was fitted using a least-squares objective function. For each time point t_i in the selected window, the model prediction $PV_{\text{model}}(t_i)$ was calculated and compared to the measured value $PV_{\text{meas}}(t_i)$. The sum of squared errors was defined as

$$S(K_p, \tau, \theta) = \sum_{i=1}^N (PV_{\text{meas}}(t_i) - PV_{\text{model}}(t_i))^2$$

This objective function measures the total mismatch between the model and the plant data over the full window. Squaring the error prevents positive and negative deviations from canceling and gives greater weight to larger mismatches.

For each disturbance, the same procedure was followed. First, the step time t_s was identified directly from the manipulated-variable record as the point where the input began leaving its original plateau.

Next, the average input before and after the step was used to calculate ΔMV . The average output before the response was used as PV_0 , while the average output after the response had mostly settled was used as PV_f . These values gave an initial estimate of the gain.

The dead time θ was interpreted physically as the delay between the input change and the start of the sustained output response. The time constant τ was interpreted as the speed at which the process approached its new steady state after the dead time had passed. Initial estimates of θ and τ were chosen from the visible shape of the response and then refined with Solver through SSE minimization.

The fitted parameters for each disturbance can be summarized in the following table.

Disturbance	Time Window	ΔMV	K_p	θ	τ
Fitting window	12:52:12 to 12:56:26	2	3.26	32	12
Downstep	12:56:28 to 13:00:00	-2	2.89	12.2	20.4
Upstep	13:00:40 to 13:04:52	2	2.92	14.7	18.3
Downstep 2	13:04:52 to end	-2	2.68	8.5	24.8
Average	-	-	2.94	16.85	18.88

After fitting each disturbance, one representative set of process parameters was obtained by averaging the individual disturbance fits:

$$K_p = \frac{K_{p,1} + K_{p,2} + K_{p,3} + K_{p,4}}{4}$$

$$\tau = \frac{\tau_1 + \tau_2 + \tau_3 + \tau_4}{4}$$

$$\theta = \frac{\theta_1 + \theta_2 + \theta_3 + \theta_4}{4}$$

If only the three later disturbance fits are used in the average, then the denominator should be changed accordingly. The final representative process model is therefore

$$G_p(s) = \frac{K_p e^{-\theta s}}{\tau s + 1}$$

Substituting the final fitted values gives

$$G_p(s) = \frac{2.94 e^{-16.85s}}{18.88s + 1}$$

Once the representative parameters were selected, the model could be simulated over the full dataset to compare predicted and measured process-variable behavior. The dead time was implemented by delaying the manipulated-variable signal:

$$MV_d(t) = MV(t - \theta)$$

In discrete form, with sampling time Δt , the delay corresponds to

$$n_\theta = \text{round}\left(\frac{\theta}{\Delta t}\right)$$

The delayed input was then used in a first-order dynamic update. In deviation form,

$$\begin{aligned}\Delta MV_d(t) &= MV_d(t) - MV_0 \\ \Delta PV(t) &= PV(t) - PV_0\end{aligned}$$

and the first-order dynamics are

$$\tau \frac{d(\Delta PV)}{dt} + \Delta PV = K_p \Delta MV_d$$

A discrete update can then be written as

$$\Delta PV_k = \alpha \Delta PV_{k-1} + (1 - \alpha) K_p \Delta MV_{d,k}$$

with

$$\alpha = e^{-\Delta t/\tau}$$

Finally, the model prediction in engineering units is reconstructed as

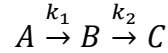
$$PV_{\text{model},k} = PV_0 + \Delta PV_k$$

This simulation allows the representative model to be compared against the entire measured record, rather than only one isolated step.

Although the model did not reproduce every fluctuation in the measured data, this was expected because the commercial plant data contained significant noise and likely included unmeasured disturbances and higher-order effects. Even with these limitations, the FOPDT model provided a reasonable engineering approximation of the dominant process dynamics. This makes the final transfer function useful for control analysis, simulation, and future controller tuning, while also showing that the identified parameters should be interpreted as best-fit empirical values rather than exact physical constants.

Problem B: Process Control Loops and Measurement Devices for Split Range Control

We begin by writing the reaction scheme for the batch reactor. The desired product is species B , and it is formed through consecutive reactions



with temperature-dependent rate expressions

$$r_1 = k_{10} \exp \left(-\frac{E_1}{RT_{R,abs}} \right) C_A$$
$$r_2 = k_{20} \exp \left(-\frac{E_2}{RT_{R,abs}} \right) C_B$$

where the absolute reactor temperature is written as

$$T_{R,abs} = T_R + 459.67$$

The assignment gives $k_{10} = 729.55 \text{ min}^{-1}$, $k_{20} = 6567.6 \text{ min}^{-1}$, $E_1 = 15000 \text{ Btu/lbmol}$, $E_2 = 20000 \text{ Btu/lbmol}$, $\Delta H_1 = -40000 \text{ Btu/lbmol}$, and $\Delta H_2 = -50000 \text{ Btu/lbmol}$. Since the process is batch, there are no inlet or outlet material streams for the reacting species, so the species balances depend only on reaction. The material balances are therefore written as:

$$\frac{dC_A}{dt} = -r_1$$
$$\frac{dC_B}{dt} = r_1 - r_2$$
$$\frac{dC_C}{dt} = r_2$$

Next, we write the reactor energy balance. The reactor gains energy from the two exothermic reactions and exchanges heat with the metal wall. This gives:

$$\rho_R V_R C_{p,R} \frac{dT_R}{dt} = (-\Delta H_1) V_R r_1 + (-\Delta H_2) V_R r_2 - U_{RM} A_{RM} (T_R - T_M)$$

or

$$\frac{dT_R}{dt} = \frac{(-\Delta H_1) V_R r_1 + (-\Delta H_2) V_R r_2 - U_{RM} A_{RM} (T_R - T_M)}{\rho_R V_R C_{p,R}}$$

This balance is used in all phases of the process. The assignment also states that the reactor metal wall must be included, so the wall temperature must be treated as its own dynamic variable. The given heat-transfer coefficients are in hours, but the simulation is run in minutes, so they are converted before being used in the model. That gives $U_{RM} = 160/60 = 2.667$, $U_{MW} = 400/60 = 6.667$, and $U_{MS} = 1000/60 = 16.667$. The given initial conditions are $C_{A0} = 0.80 \text{ lbmol/ft}^3$, $C_{B0} = 0$, $C_{C0} = 0$, and $T_R = T_M = T_J = 80^\circ\text{F}$. The reactor, wall, and jacket volumes are 42.5, 9.42, and 18.83 ft^3 , respectively.

We then write the measurement and control equations. The thermocouple has a first-order lag with time constant $\tau_s = 1 \text{ min}$. The temperature transmitter range is 50°F to 250°F mapped to 3 to 15psi, so the steady transmitter signal is:

$$TT^* = 3 + \frac{12}{200}(T_R - 50)$$

which simplifies to,

$$TT^* = 0.06T_R$$

The transmitter dynamics are therefore:

$$\frac{dTT}{dt} = \frac{TT^* - TT}{\tau_s}$$

The controller output is given in the assignment as

$$CO = 7 + K_c(SP - TT) + \frac{1}{\tau_I} \int_0^t (SP - TT) dt$$

with set point $SP = 12$ psi and controller bias equal to 7psi. Since $TT^* = 0.06T_R$, the set point of 12 psi corresponds to a reactor temperature of 200°F. For proportional-only control, the integral term is removed by taking $\tau_I \rightarrow \infty$. For PI control, it is convenient to define an additional state variable

$$IE = \int_0^t (SP - TT) dt \rightarrow$$

$$\frac{dIE}{dt} = SP - TT$$

and the final controller equation becomes

$$CO = 7 + K_c(SP - TT) + \frac{1}{\tau_I} IE$$

The initial transmitter signal at 80°F is $TT_0 = 0.06(80) = 4.8$ psi.

The split-range valve equations are written next. For the steam valve,

$$x_S = \begin{cases} 0, & CO \leq 9 \\ \frac{CO - 9}{6}, & 9 < CO < 15 \\ 1, & CO \geq 15 \end{cases}$$

For the cooling-water valve,

$$x_W = \begin{cases} 1, & CO \leq 3 \\ \frac{9 - CO}{6}, & 3 < CO < 9 \\ 0, & CO \geq 9 \end{cases}$$

The cooling-water flow rate is

$$Q_W = C_{v,W} x_W \sqrt{\Delta P_W}$$

and the water mass flow rate is

$$W_W = Q_W \left(\frac{\rho_W}{7.4805} \right)$$

These equations determine when the system is heating and when it is cooling. If $CO > 9\text{psi}$, steam is active. If $CO < 9\text{psi}$, cooling water is active.

The problem is then solved in three phases. Phase A is steam heating. Phase B is jacket filling with cooling water. Phase C is operation with the jacket completely full of water. This matches the three operating regions required in the assignment.

The differential equations are solved in the CHE Calculator with the fifth-order Runge-Kutta method over the 60 min batch time. The final conditions from one phase are used as the initial conditions for the next phase, and the time axes are combined into one continuous temperature plot.

For the proportional-only study, the integral term is removed by setting $\tau_I = \infty$, and the model is solved for $K_c = 2, 4, 8$, and 10 .

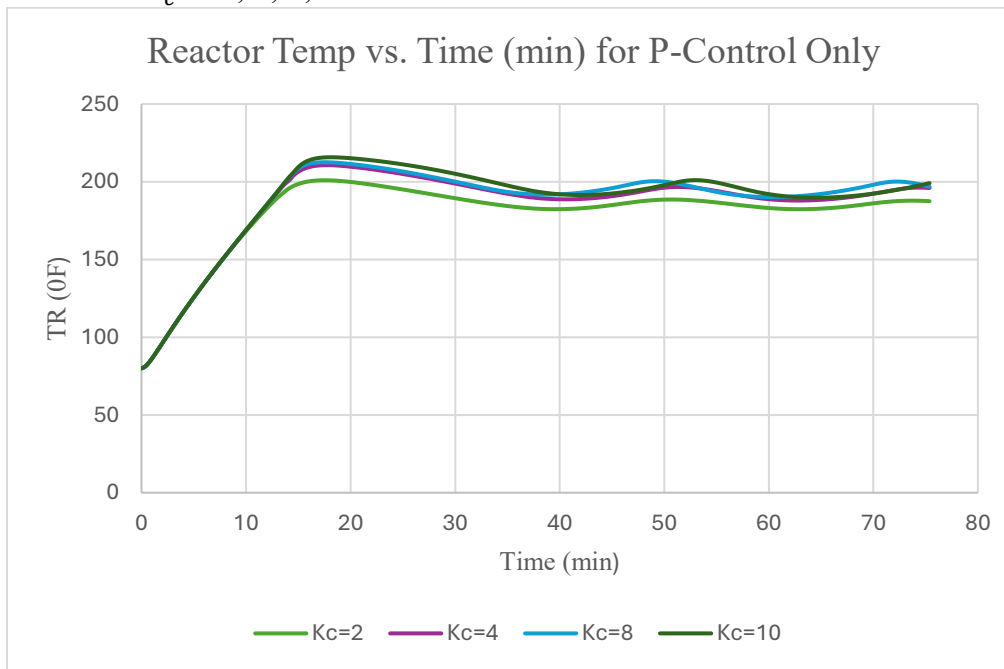


Figure 1: Reactor temperature versus time for proportional-only control with $K_c = 2, 4, 6, 8$, and $\tau_I = \infty$

As K_c increases, the controller becomes more aggressive. The temperature moves closer to the set point, but the response becomes less smooth and more oscillatory. Lower K_c gives smoother behavior but larger offset. Higher K_c reduces offset somewhat, but it increases overshoot. This is the expected tradeoff for proportional control.

To reduce the offset, PI control is then tested. A convenient way to do this is to first hold $K_c = 2$ and compare several reset times, such as $\tau_I = 20, 30, 40$, and 50min .

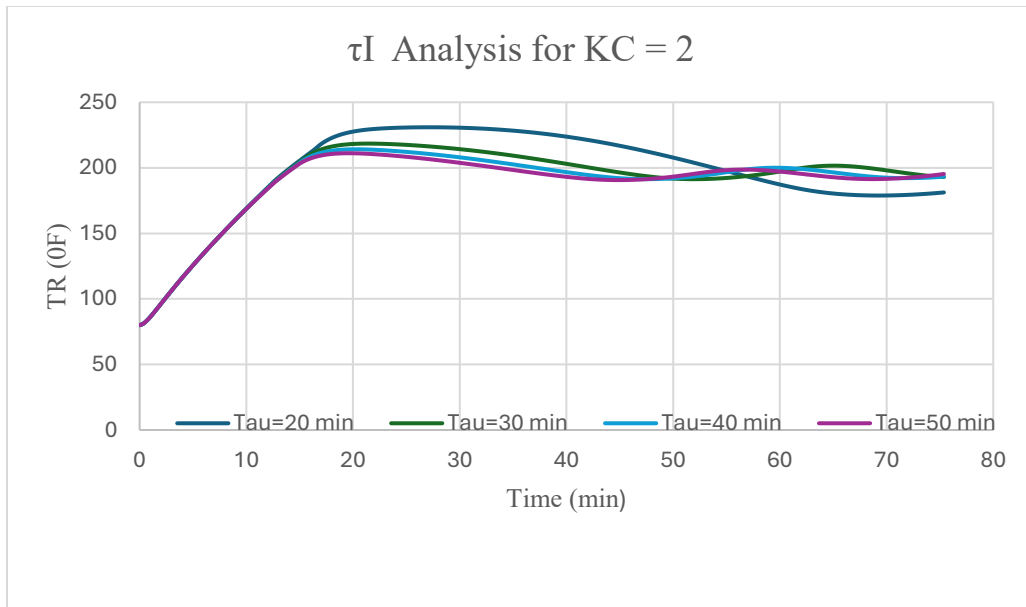


Figure 2: Reactor temperature versus time for $\tau_I = 20, 30, 40, 50$ min at fixed $K_c = 2$

Next, with $\tau_I = 40$ min fixed, the gain is varied again for $K_c = 2, 4, 8$, and 10 . The PI results show that the offset is reduced compared with proportional-only control, but high gain still causes more oscillation. Among the tested values, $K_c = 4$ gives the best overall response.

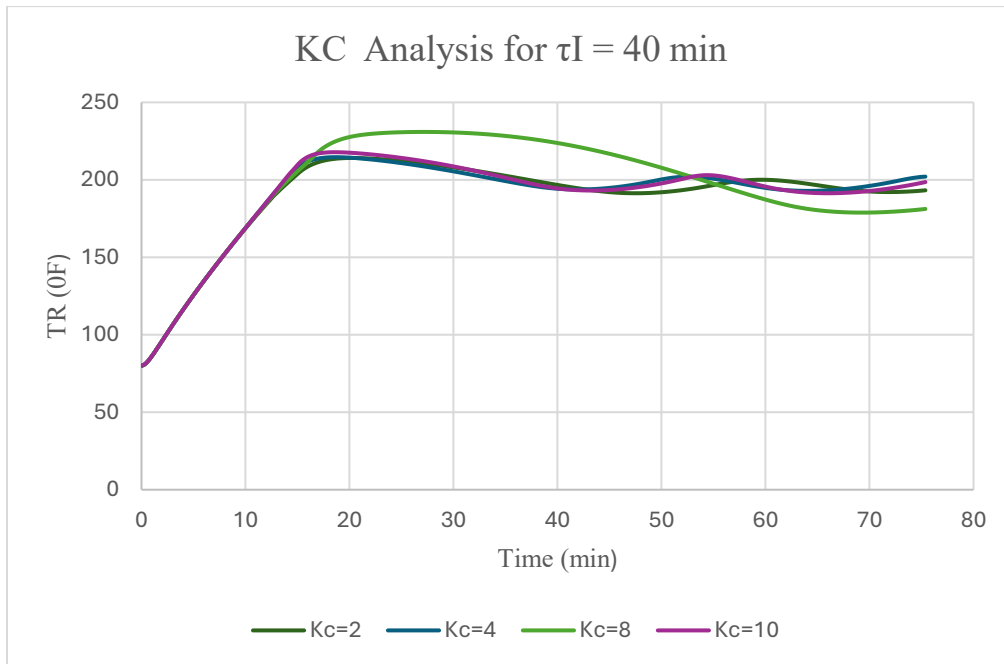


Figure 3: Reactor temperature versus time for the selected PI tuning $K_c = 4$ and $\tau_I = 40$ min

The final recommended controller settings are

$$K_c = 4, \tau_I = 40 \text{ min}$$

These values give a good compromise between offset removal and smooth control action.

Appendix I: ODE Code

Function fODE_A(T As Double, u() As Double, Optional Param As Range)

Dim n As Integer

n = UBound(u)

ReDim f(1 To n) As Double

' Read all model and controller parameters from the worksheet

Dim k10 As Double, k20 As Double, E1 As Double, E2 As Double

Dim dH1 As Double, dH2 As Double, Rgas As Double

Dim VR As Double, VM As Double, ARM As Double, AMJ As Double

Dim rhoR As Double, rhoM As Double

Dim CpR As Double, CpM As Double

Dim URM As Double, UMS As Double

Dim tauS As Double, SP As Double, bias As Double

Dim Kc As Double, invTauI As Double

Dim Tsat35 As Double

k10 = Param.Cells(1, 1)

k20 = Param.Cells(2, 1)

E1 = Param.Cells(3, 1)

E2 = Param.Cells(4, 1)

dH1 = Param.Cells(5, 1)

dH2 = Param.Cells(6, 1)

Rgas = Param.Cells(7, 1)

VR = Param.Cells(8, 1)

VM = Param.Cells(9, 1)

ARM = Param.Cells(11, 1)

AMJ = Param.Cells(12, 1)

rhoR = Param.Cells(13, 1)

rhoM = Param.Cells(14, 1)

CpR = Param.Cells(16, 1)

CpM = Param.Cells(17, 1)

URM = Param.Cells(19, 1)

UMS = Param.Cells(21, 1)

tauS = Param.Cells(22, 1)

SP = Param.Cells(23, 1)

bias = Param.Cells(24, 1)

Kc = Param.Cells(25, 1)

invTauI = Param.Cells(26, 1)

Tsat35 = Param.Cells(32, 1)

' Unpack the state variables

Dim CA As Double, CB As Double, CC As Double

Dim TR As Double, TM As Double, TT As Double

Dim MJ As Double, TJ As Double, IE As Double

CA = u(1)

```
CB = u(2)
CC = u(3)
TR = u(4)
TM = u(5)
TT = u(6)
MJ = u(7)
TJ = u(8)
IE = u(9)
```

```
' Prevent concentrations from becoming slightly negative during integration
```

```
If CA < 0# Then CA = 0#
```

```
If CB < 0# Then CB = 0#
```

```
If CC < 0# Then CC = 0#
```

```
' Calculate absolute temperature and reaction rates
```

```
Dim TRabs As Double
```

```
Dim r1 As Double, r2 As Double
```

```
Dim TTss As Double, err As Double
```

```
Dim CO As Double, xs As Double
```

```
Dim Qrxn As Double, QRM As Double, QMS As Double
```

```
TRabs = TR + 459.67
```

```
If TRabs < 1# Then TRabs = 1#
```

```
r1 = k10 * Exp(-E1 / (Rgas * TRabs)) * CA
```

```
r2 = k20 * Exp(-E2 / (Rgas * TRabs)) * CB
```

```
' Evaluate the transmitter and controller equations
```

```
TTss = 3# + (12# / 200#) * (TR - 50#)
```

```
err = SP - TT
```

```
CO = bias + Kc * err + invTauI * IE
```

```
If CO < 3# Then CO = 3#
```

```
If CO > 15# Then CO = 15#
```

```
' Determine steam-valve opening for split-range control
```

```
If CO <= 9# Then
```

```
    xs = 0#
```

```
Else
```

```
    xs = (CO - 9#) / 6#
```

```
End If
```

```
If xs < 0# Then xs = 0#
```

```
If xs > 1# Then xs = 1#
```

```
' Compute reaction heat, reactor-to-wall heat transfer, and steam heating
```

```
Qrxn = (-dH1) * VR * r1 + (-dH2) * VR * r2
```

```
QRM = URM * ARM * (TR - TM)
```

```
QMS = xs * UMS * AMJ * (Tsat35 - TM)
```

' Material balances on A, B, and C

$$f(1) = -r1$$

$$f(2) = r1 - r2$$

$$f(3) = r2$$

' Reactor and wall energy balances for the steam-heating phase

$$f(4) = (Q_{rxn} - Q_{RM}) / (\rho_{oR} * V_R * C_{pR})$$

$$f(5) = (Q_{RM} + Q_{MS}) / (\rho_{oM} * V_M * C_{pM})$$

' Transmitter dynamics

$$f(6) = (T_{Tss} - T_T) / \tau_{aS}$$

' In Phase A, jacket-water mass and jacket-water temperature do not change

$$f(7) = 0\#$$

$$f(8) = 0\#$$

' Integral-of-error state

$$f(9) = err$$

$$fODE_A = f$$

End Function

Function fODE_B(T As Double, u() As Double, Optional Param As Range)

Dim n As Integer

n = UBound(u)

ReDim f(1 To n) As Double

' Read all model and controller parameters from the worksheet

Dim k10 As Double, k20 As Double, E1 As Double, E2 As Double

Dim dH1 As Double, dH2 As Double, Rgas As Double

Dim VR As Double, VM As Double, ARM As Double, AMJ As Double

Dim rhoR As Double, rhoM As Double, rhoW As Double

Dim CpR As Double, CpM As Double, CpW As Double

Dim URM As Double, UMW As Double

Dim tauS As Double, SP As Double, bias As Double

Dim Kc As Double, invTauI As Double

Dim CvW As Double, dPW As Double, TinW As Double

k10 = Param.Cells(1, 1)

k20 = Param.Cells(2, 1)

E1 = Param.Cells(3, 1)

E2 = Param.Cells(4, 1)

dH1 = Param.Cells(5, 1)

dH2 = Param.Cells(6, 1)

Rgas = Param.Cells(7, 1)

```

VR = Param.Cells(8, 1)
VM = Param.Cells(9, 1)
ARM = Param.Cells(11, 1)
AMJ = Param.Cells(12, 1)
rhoR = Param.Cells(13, 1)
rhoM = Param.Cells(14, 1)
rhoW = Param.Cells(15, 1)
CpR = Param.Cells(16, 1)
CpM = Param.Cells(17, 1)
CpW = Param.Cells(18, 1)
URM = Param.Cells(19, 1)
UMW = Param.Cells(20, 1)
tauS = Param.Cells(22, 1)
SP = Param.Cells(23, 1)
bias = Param.Cells(24, 1)
Kc = Param.Cells(25, 1)
invTauI = Param.Cells(26, 1)
CvW = Param.Cells(27, 1)
dPW = Param.Cells(28, 1)
TinW = Param.Cells(31, 1)

```

' Unpack the state variables

```

Dim CA As Double, CB As Double, CC As Double
Dim TR As Double, TM As Double, TT As Double
Dim MJ As Double, TJ As Double, IE As Double

```

```

CA = u(1)
CB = u(2)
CC = u(3)
TR = u(4)
TM = u(5)
TT = u(6)
MJ = u(7)
TJ = u(8)
IE = u(9)

```

' Prevent concentrations and jacket mass from becoming unrealistically small

```

If CA < 0# Then CA = 0#
If CB < 0# Then CB = 0#
If CC < 0# Then CC = 0#
If MJ < 35# Then MJ = 35#

```

' Calculate absolute temperature and reaction rates

```

Dim TRabs As Double
Dim r1 As Double, r2 As Double
Dim TTss As Double, err As Double
Dim CO As Double, xw As Double
Dim Ww As Double

```

Dim Qrxn As Double, QRM As Double, QMW As Double

$$TRabs = TR + 459.67$$

If TRabs < 1# Then TRabs = 1#

$$r1 = k10 * \text{Exp}(-E1 / (Rgas * TRabs)) * CA$$

$$r2 = k20 * \text{Exp}(-E2 / (Rgas * TRabs)) * CB$$

' Evaluate the transmitter and controller equations

$$TTss = 3\# + (12\# / 200\#) * (TR - 50\#)$$

$$err = SP - TT$$

$$CO = bias + Kc * err + invTauI * IE$$

If CO < 3# Then CO = 3#

If CO > 15# Then CO = 15#

' Changed for Phase B: steam heating is removed, cooling-water flow is activated,

' jacket-water mass now increases, and jacket-water temperature becomes dynamic

If CO <= 3# Then

$$xw = 1\#$$

ElseIf CO < 9# Then

$$xw = (9\# - CO) / 6\#$$

Else

$$xw = 0\#$$

End If

$$Ww = CvW * xw * \text{Sqr}(dPW) * (\rho W / 7.4805)$$

$$Qrxn = (-dH1) * VR * r1 + (-dH2) * VR * r2$$

$$QRM = URM * ARM * (TR - TM)$$

$$QMW = UMW * AMJ * (TM - TJ)$$

' Material balances on A, B, and C

$$f(1) = -r1$$

$$f(2) = r1 - r2$$

$$f(3) = r2$$

' Reactor and wall energy balances for the jacket-filling phase

$$f(4) = (Qrxn - QRM) / (\rho R * VR * CpR)$$

$$f(5) = (QRM - QMW) / (\rho M * VM * CpM)$$

' Transmitter dynamics

$$f(6) = (TTss - TT) / \tau S$$

' Jacket-water mass and energy balances during filling

$$f(7) = Ww$$

$$f(8) = (Ww * CpW * (TinW - TJ) + QMW) / (MJ * CpW)$$

' Integral-of-error state

f(9) = err

fODE_B = f

End Function

Function fODE_C(T As Double, u() As Double, Optional Param As Range)

Dim n As Integer

n = UBound(u)

ReDim f(1 To n) As Double

' Read all model and controller parameters from the worksheet

Dim k10 As Double, k20 As Double, E1 As Double, E2 As Double

Dim dH1 As Double, dH2 As Double, Rgas As Double

Dim VR As Double, VM As Double, VJ As Double, ARM As Double, AMJ As Double

Dim rhoR As Double, rhoM As Double, rhoW As Double

Dim CpR As Double, CpM As Double, CpW As Double

Dim URM As Double, UMW As Double

Dim tauS As Double, SP As Double, bias As Double

Dim Kc As Double, invTauI As Double

Dim CvW As Double, dPW As Double, TinW As Double

k10 = Param.Cells(1, 1)

k20 = Param.Cells(2, 1)

E1 = Param.Cells(3, 1)

E2 = Param.Cells(4, 1)

dH1 = Param.Cells(5, 1)

dH2 = Param.Cells(6, 1)

Rgas = Param.Cells(7, 1)

VR = Param.Cells(8, 1)

VM = Param.Cells(9, 1)

VJ = Param.Cells(10, 1)

ARM = Param.Cells(11, 1)

AMJ = Param.Cells(12, 1)

rhoR = Param.Cells(13, 1)

rhoM = Param.Cells(14, 1)

rhoW = Param.Cells(15, 1)

CpR = Param.Cells(16, 1)

CpM = Param.Cells(17, 1)

CpW = Param.Cells(18, 1)

URM = Param.Cells(19, 1)

UMW = Param.Cells(20, 1)

tauS = Param.Cells(22, 1)

SP = Param.Cells(23, 1)

bias = Param.Cells(24, 1)

Kc = Param.Cells(25, 1)

invTauI = Param.Cells(26, 1)

```
CvW = Param.Cells(27, 1)
dPW = Param.Cells(28, 1)
TinW = Param.Cells(31, 1)
```

```
' Unpack the state variables
Dim CA As Double, CB As Double, CC As Double
Dim TR As Double, TM As Double, TT As Double
Dim MJ As Double, TJ As Double, IE As Double
```

```
CA = u(1)
CB = u(2)
CC = u(3)
TR = u(4)
TM = u(5)
TT = u(6)
MJ = u(7)
TJ = u(8)
IE = u(9)
```

```
' Prevent concentrations from becoming slightly negative during integration
If CA < 0# Then CA = 0#
If CB < 0# Then CB = 0#
If CC < 0# Then CC = 0#
```

```
' Calculate absolute temperature and reaction rates
Dim TRabs As Double
Dim r1 As Double, r2 As Double
Dim TTss As Double, err As Double
Dim CO As Double, xw As Double
Dim Ww As Double
Dim Qrxn As Double, QRM As Double, QMW As Double
Dim Mfull As Double
```

```
TRabs = TR + 459.67
If TRabs < 1# Then TRabs = 1#
```

```
r1 = k10 * Exp(-E1 / (Rgas * TRabs)) * CA
r2 = k20 * Exp(-E2 / (Rgas * TRabs)) * CB
```

```
' Evaluate the transmitter and controller equations
TTss = 3# + (12# / 200#) * (TR - 50#)
```

```
err = SP - TT
CO = bias + Kc * err + invTauI * IE
If CO < 3# Then CO = 3#
If CO > 15# Then CO = 15#
```

```
' Changed for Phase C: the jacket is now full, so jacket mass is fixed
```

' and the jacket-water energy balance uses the constant full-jacket mass

If CO <= 3# Then

$$xw = 1\#$$

ElseIf CO < 9# Then

$$xw = (9\# - CO) / 6\#$$

Else

$$xw = 0\#$$

End If

$$Ww = CvW * xw * Sqr(dPW) * (rhoW / 7.4805)$$

$$Qrxn = (-dH1) * VR * r1 + (-dH2) * VR * r2$$

$$QRM = URM * ARM * (TR - TM)$$

$$QMW = UMW * AMJ * (TM - TJ)$$

$$Mfull = rhoW * VJ$$

' Material balances on A, B, and C

$$f(1) = -r1$$

$$f(2) = r1 - r2$$

$$f(3) = r2$$

' Reactor and wall energy balances for the full-jacket phase

$$f(4) = (Qrxn - QRM) / (rhoR * VR * CpR)$$

$$f(5) = (QRM - QMW) / (rhoM * VM * CpM)$$

' Transmitter dynamics

$$f(6) = (TTss - TT) / tauS$$

' Jacket mass is fixed, and jacket-water temperature changes using the full-jacket mass

$$f(7) = 0\#$$

$$f(8) = (Ww * CpW * (TinW - TJ) + QMW) / (Mfull * CpW)$$

' Integral-of-error state

$$f(9) = err$$

$$fODE_C = f$$

End Function

References

1. Chat GPT
2. Michael Caracotsios, *Introduction to Chemical Process Control*, 2026
3. Chemical Engineering Calculator v13